

Hybrid Gains and Tasty Grains: Comparing Production- and Consumption-Focused Seed Promotion Strategies to Accelerate Varietal Turnover

Bjorn Van Campenhout*, Leocardia Nabwire†, Berber Kramer‡,
Carly Trachtman§, Gashaw T. Abate§

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Abstract

Accelerating varietal turnover—replacing older crop varieties with newer, higher-yielding and more resilient ones—is a central goal of seed-system policy in sub-Saharan Africa, yet how best to promote it remains unclear. The fact that smallholders are generally both producers and consumers suggest that seed promotion strategies should emphasize both agronomic performance as well as consumption related traits. In a cluster-randomized trial, we compare two seed promotion strategies among maize-growing households in eastern Uganda: seed trial packs, which are intended to emphasize farmers’ learning about production traits through on-farm experimentation, and cooking demonstrations with blind tasting, which are intended to emphasize learning about consumption traits through direct experience. Both raise farmers’ knowledge and use of the promoted hybrid. Trial packs generate the larger increase, but it runs almost entirely through farmers recycling grain saved from the trial pack rather than buying fresh seed, a rational choice in a context of low grain prices, high seed prices, uncertain quality, and liquidity constraints. Cooking demonstrations produce a smaller increase that runs through fresh-seed purchases, with suggestive yield gains. Trial packs are more cost-effective at the scale studied, but their relative advantage depends both on the technology and the delivery context. For recyclable varieties such as Open Pollinated Varieties (OPVs), they directly accelerate varietal turnover. For hybrids, their long-run impact depends on farmers returning to the seed market for fresh seed. Moreover, because cooking demonstrations have largely fixed delivery costs, they become increasingly competitive when they can be delivered to larger groups.

Keywords: technology adoption, consumption and production traits, trial packs, cooking demonstration and blind tasting, maize, Uganda.

JEL: Q16, O33, Q12, D83, C93

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*Markets, Trade, and Institutions Unit, International Food Policy Research Institute, Leuven, Belgium, corresponding author: b.vancampenhout@cgiar.org

†One Acre Fund, Jijina, Uganda

‡Markets, Trade, and Institutions Unit, International Food Policy Research Institute, Nairobi, Kenya

§Markets, Trade, and Institutions Unit, International Food Policy Research Institute, Washington DC, USA

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